

Unit 1 Assessment, Part 2

1. Use the characteristics of the given function to graph the function and its reciprocal on the same set of axes, either using online graphing technology or by hand. Find and state all aspects needed to sketch the function and its reciprocal, where possible. Explain your thinking process. **[K6] [C2]**

$$f(x) = -x^2 + 7x - 6$$



2. a. Fill in the table and sketch the graph of the following equation. **[A10], [C2]**

$$f(x) = \frac{-2x - 5}{3x + 18}$$

Vertical asymptotes	Horizontal asymptotes	x– intercept	y– intercept	Domain



- 2 b. State positive and negative intervals for $f(x) = \frac{-2x-5}{3x+18}$ [K2]

Positive interval:

Negative intervals:

3. Find the real roots of the following rational equations. [K8] [C2]

a. $\frac{-7x}{9x+11} - 12 = \frac{1}{x}$

b. $\frac{x-1}{x+2} = \frac{3x+8}{5x-1}$

4. Solve the following inequality algebraically. Explain your process. [A4] [C2]

$$8x - 3 \leq 2x + 1 \leq 17x - 8$$

5. Solve the following inequality algebraically. Include an interval chart in your solution.

[A6] [C2]

$$\frac{5x+4}{x-11} < \frac{5x-7}{x+13}$$

6. The dimensions of a box are 3 cm × 5 cm × 7 cm. The width, length, and height of the box must be increased by the same amount X in order for the box to have a volume of 693 cm³. Determine the value of X that will produce a box with a volume of 693 cm³. **[T4] [C2]**

7. The specifications for a storage container state that the length is 1 metre more than triple the width and the height is 5 metres less than double the width. Find the range of possible dimensions for a volume of at least 8436 m^3 using the algebraic method. **[T6] [C2]**